

Foundation (Jalapeno)

- Q1.** What is the purpose of graphing multiple inequalities on the same coordinate plane?
- (A) To find the intersection point
 (B) To identify overlapping solution regions
 (C) To determine the slope of a line
 (D) To find the equation of a line
 (E) To calculate the area of a triangle
- Q2.** Which of the following is a feasible region for the system of inequalities $x + y \geq 2$ and $x - y \leq 1$?
- (A) The region above the line $x + y = 2$ and below the line $x - y = 1$
 (B) The region below the line $x + y = 2$ and above the line $x - y = 1$
 (C) The region above the line $x + y = 2$ and above the line $x - y = 1$
 (D) The region below the line $x + y = 2$ and below the line $x - y = 1$
 (E) The region on the line $x + y = 2$ and on the line $x - y = 1$
- Q3.** What is the relationship between the graphs of $x \geq 3$ and $x \leq 5$?
- (A) They are parallel lines
 (B) They are perpendicular lines
 (C) They are overlapping solution regions
 (D) They are intersecting lines
 (E) They are asymptotes
- Q4.** How many solution regions are formed by the system of inequalities $y \geq x$ and $y \leq -x$?
- (A) 1
 (B) 2
 (C) 3
 (D) 4
 (E) Infinitely many
- Q5.** Which of the following systems of inequalities has no solution?
- (A) $x \geq 2$ and $x \leq 1$
 (B) $x \geq 1$ and $x \leq 2$
 (C) $x \geq 3$ and $x \leq 4$
 (D) $x \geq 5$ and $x \leq 6$
 (E) $x \geq 7$ and $x \leq 8$
- Q6.** What is the purpose of identifying overlapping solution regions in a system of linear inequalities?
- (A) To find the equation of a line
 (B) To calculate the area of a triangle
 (C) To determine the feasible region
 (D) To find the intersection point
 (E) To graph a line
- Q7.** Which of the following is a correct method for graphing the inequality $x + y \geq 3$?
- (A) Graph the line $x + y = 3$ and shade the region above it
 (B) Graph the line $x + y = 3$ and shade the region below it
 (C) Graph the line $x + y = 3$ and shade the region on it
 (D) Graph the line $x + y = 3$ and shade the region to the left of it
 (E) Graph the line $x + y = 3$ and shade the region to the right of it
- Q8.** How many lines are formed by the system of inequalities $x \geq 2$ and $y \geq 3$?
- (A) 1
 (B) 2
 (C) 3
 (D) 4
 (E) Infinitely many

Open Ended Questions (FRQ)

- Q9.** Graph the system of inequalities $x + y \geq 2$ and $x - y \leq 1$. Identify the overlapping solution regions and interpret the feasible region.
- Q10.** Solve the system of inequalities $y \geq x$ and $y \leq -x$. Graph the solution regions and identify the points of intersection.

Chef's Tips

- Tip 1: Encourage students to use graph paper to visualize the solution regions
- Tip 2: Remind students to label the axes and title the graph
- Tip 3: Emphasize the importance of checking the solution regions for overlap and feasibility